

FileEditSelectionViewGoRunTerminalHelp

Assignment 12

ASSIGNME...cv.ipynb

Jupyter NotebookPython 3.11.9

C: > TuteDude > Data Science Combo > Python > Assignment 11 > cv.ipynb > import cv2

Clear All OutputsGo ToJupyter VariablesOutline

Displayed image



image.  
  
, image)  
  
age)  
lly as Output.jpg")  
  
pg found, make sure it is named input.jpg and placed in the current directory before trying again!')

Python

method for resizing the image by entering the corresponding number.\n1. Exact dimensions\n2. Downscale to a percentage  
  
choice = input('Enter your choice: ').strip()  
  
if not choice.isdigit():  
 print('Please enter a valid number!')

Python

Spaces: 4 {} Cell 2 of 24

 Code
  Markdown
 | 
  Interrupt
  Restart
  Clear All

```
except ValueError:
    print('Please only
width = None
height = None
```

```
cv2.imshow('Original Image', image)
cv2.imshow('Resized Image', res_image)
```

```
cv2.waitKey(0)
cv2.destroyAllWindows()
cv2.waitKey(1)
```

```
cv2.imwrite("resized_image.jpg", res_image)
print('Image successfully saved as resized')
```

[17] 11.3s

```

Please choose the method for resizing the image
1. Exact dimensions
2. Downscale to a percentage of original size
3. Scale down using INTER_AREA (Best quality for
Enter 0 to go back

```

## Flipping images (horizontal, vertical, both)

```
import cv2
```

|     |   |      |
|-----|---|------|
| [2] | ✓ | 5.1s |
|-----|---|------|



 Original Image



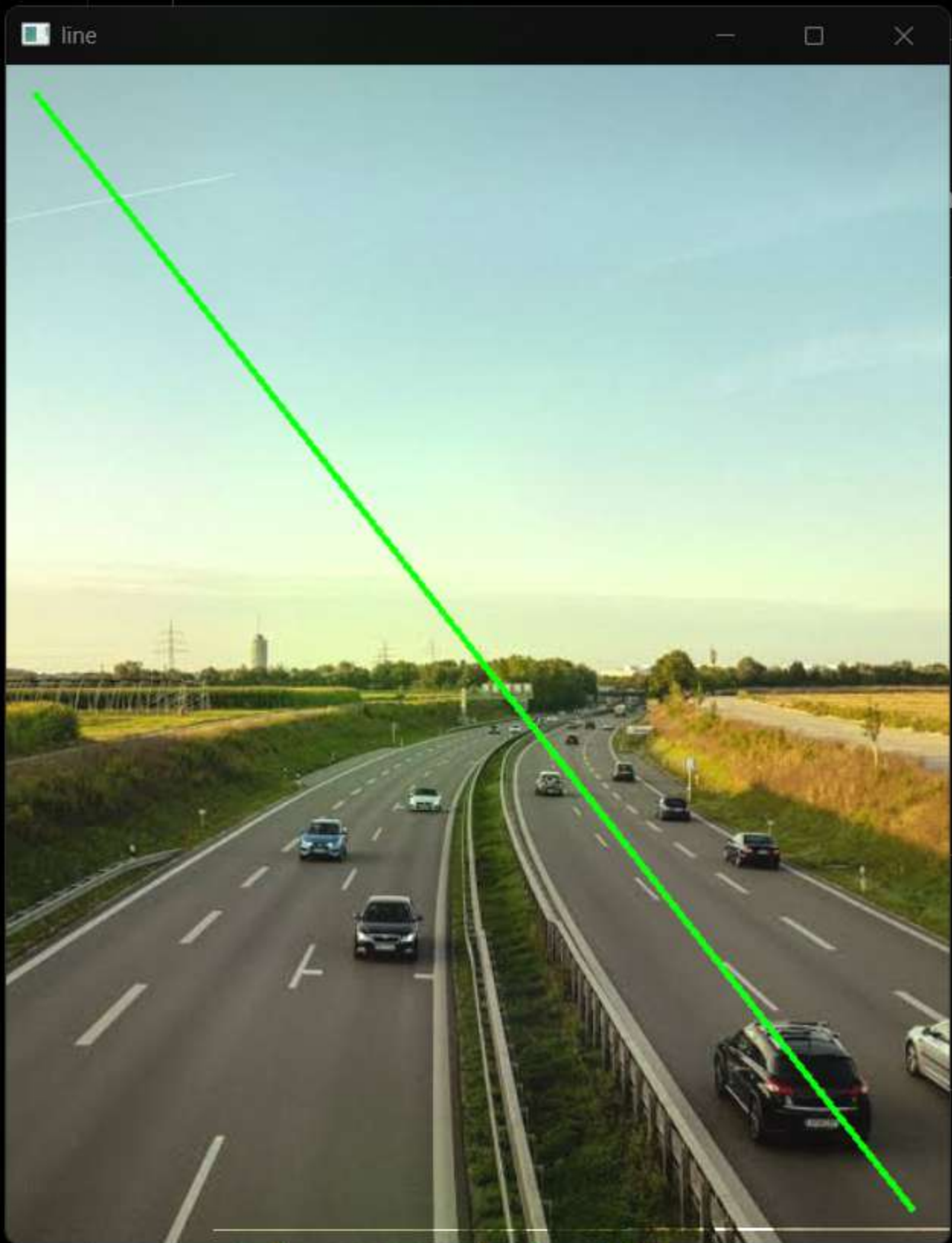
$(0, 255, 0), 2)$   
 $(500), (0, 255, 0), 2)$

 **MICROSOFT RESEARCH**

Python

Python

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+ Code + Markdown | Interrupt Restart Clear All Outputs Go To Jupyter Variables Outline



```
height, width = img.shape[:2]
```

[4] ✓ 1.5s

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cv.ipynb

Python 3.11.9

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```
cv2.waitKey(0)
cv2.destroyAllWindows()
print('Executed Successfully')
```

Executed Successfully

import cv2
img = cv2.imread('input.jpg')
text = input('Enter text: ')

cv2.putText(
 img, text, (50, 50), cv2.FONT\_HERSHEY\_SIMPLEX, 1, (0, 255, 0), 2
)

cv2.imshow('Result', img)
cv2.waitKey(0)
cv2.destroyAllWindows()

13.4s

Image translation (shifting) using warpAffine

[26]

```
import cv2
import numpy as np

img = cv2.imread('input.jpg')
height, width = img.shape[:2]
tx = input('Enter for horizontal')
ty = input("Enter for veritcal")

M = np.float32([
    [1  0  tx]
    [0  1  ty]
```

Result



Python

Spaces: 4LF{}Cell 9 of 25

FileEditSelectionViewGoRunTerminalHelpAssignment 12

Translated



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RestartClear All OutputsGo ToJupyter VariablesOutline

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Python

using warpAffine

```
img = cv.imread('img.jpg')
img = cv.resize(img, (width, height),
                interpolation=cv.INTER_LINEAR)
img = cv.cvtColor(img, cv.COLOR_BGR2GRAY)
img = cv.warpAffine(img, M, (width, height))
cv.imshow('translated_img', translated_img)
cv.waitKey(0)
cv.destroyAllWindows()
print('Executed Successfully')
```

[26] 7.6s

Python

Image rotation using getRotationMatrix2D and warpAffine

Spaces: 4 {} Cell 10 of 24

FileEditSelectionViewGoRunTerminalHelp

Assignment 12

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Python 3.11.9

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+ Code+ MarkdownInterruptRestartClear All OutputsGo ToJupyter VariablesOutline

[26] ✓ 11.2s

Rotated



nd warpAffine

wise, negative = clockwise

gle, scale)

width, height))

Python

[6] ✓ 3.3s

img = cv2.imread('input\_img', cv2.IMREAD\_GRAYSCALE)

Python

Spaces: 4 {} Cell 12 of 24

Navigation icons: Home, Search, Recent, View, Run, Clear All Outputs, Restart, Python 3.11.9, Python



```
on > Assignment 11 > cv.ipynb >
Restart Clear All Outputs
lly')
```

```
og', cv2.IMREAD_GRAYSCALE)
threshold(img, thresh_value,
cv2.threshold(img, thresh_valu
ry_thresh)
ed', binary_inv_thresh)
lly')
```

Blurring images using Gaussian Blur and Median Blur

```
import cv2
```

[7] ✓ 4.2s Python

FileEditSelectionViewGoRunTerminalHelp

Assignment 12

ASSIGNME...cv.ipynb

Gaussian Blur



Assignment 11>cv.ipynb>import cv2

RestartClear All OutputsGo ToJupyter VariablesOutline

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Python

Blur and Median Blur

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')

# Gaussian Blur
gaussian_blur = cv2.GaussianBlur(img, (7, 7), 0)

# Median Blur
median_blur = cv2.medianBlur(img, 5)

cv2.imshow('Gaussian Blur', gaussian_blur)
cv2.imshow('Median Blur', median_blur)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Python

Morphological operations (Tophat, Blackhat)

```
import cv2
import numpy as np
```

[8] ✓ 4.3s

Python

Spaces: 4 {} Cell 16 of 24

FileEditSelectionViewGoRunTerminalHelp

Assignment 12

Median Blur



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RestartClear All OutputsGo ToJupyter VariablesOutlinePython 3.11.9

Python

Median Blur and Median Blur

```
img = cv.imread('img.jpg')
img = cv.GaussianBlur(img, (7, 7), 0)
cv.imwrite('gaussian_blur.jpg', gaussian_blur)
img = cv.imread('img.jpg')
img = cv.medianBlur(img, 5)
cv.imwrite('median_blur.jpg', median_blur)
cv.imwrite('successfully')
```

Python

Morphological operations (Tophat, Blackhat)

```
import cv2
import numpy as np
```

[8] ✓ 4.3s

Python

Spaces: 4 {} Cell 16 of 24

FileEditSelectionViewGoRunTerminalHelp

Assignment 12

Original

Top-Hat (Bright details)

Black-Hat (Dark details)

Python

Python

Python

Edge detection using Canny

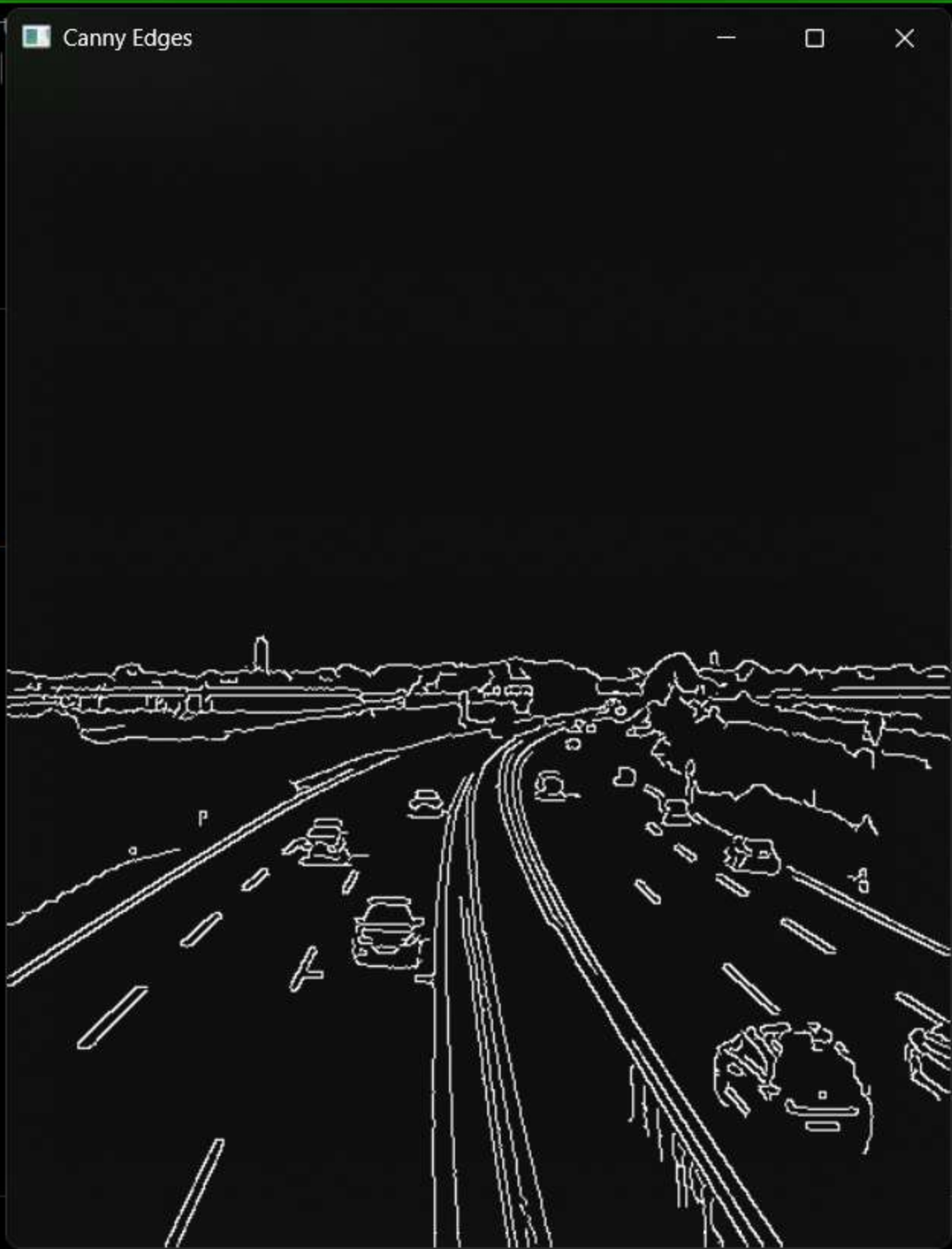
Spaces: 4 { } Cell 18 of 24

[30] 8.7s

Edge detection using Canny



```
Python > Assignment 11 > cv.ipynb > import  
t Restart Clear All Outputs Go To  
(light_details), topmat,  
(Dark_details)', blackhat)  
)  
sfully')  
  
t.jpg', cv2.IMREAD_GRAYSCALE)  
Blur(img, (5, 5), 1.4)  
  
red, threshold1, threshold2)  
img)  
s', edges)  
)  
sfully')
```



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Python 3.11.9  
Python

Reading and writing video files

```
import cv2
```

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Assignment 12

Video Stream



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Python

```
4')

file")

_FRAME_WIDTH))
P_FRAME_HEIGHT))

*'mp4v')
4', fourcc, fps, (width, height))

or(frame, cv2.COLOR_BGR2GRAY)
tColor(processed_frame, cv2.COLOR_GRAY2BGR)

r)

rocessed_frame_bgr)

ord('q'):
```

Spaces: 4 {} Cell 22 of 24

